

MSPA – 4

Area, Power, and Performance Trade-Off Analysis

Course: Synthesis and Optimization of VLSI Circuits

Tools: Vivado Design Suite, Verilog HDL

Date of Submission: 09-04-2026

Objective

The objective of this assignment is to analyze the trade-offs between **area, power, and performance** in digital circuit design. Students will understand how different synthesis strategies affect hardware utilization, power consumption, and speed of the design.

Instructions

- Write synthesizable Verilog HDL only
 - Avoid delays (#) and non-synthesizable constructs
 - Create a testbench for verification
 - Perform synthesis and implementation
 - Apply optimization strategies
 - Generate reports for **area, power, and timing**
 - Compare results and record observations
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Problem 1: Baseline Design

Task

Design a **4-bit ALU** supporting:

- Addition
- Subtraction
- AND
- OR

Steps

- Write Verilog code
- Simulate using testbench
- Perform synthesis (default settings)
- Generate reports

Record

- LUT count (Area)
- Power consumption
- Critical path delay
- Maximum frequency

Problem 2: Area Optimization

Task

Apply **area optimization strategy** in synthesis

Steps

- Use synthesis directive: Area Optimized
- Run synthesis and implementation

Observe

- Reduction in LUTs/registers
- Impact on delay

- Change in frequency

Problem 3: Performance Optimization

Task

Apply **performance optimization strategy**

Steps

- Use performance-oriented synthesis
- Re-run implementation

Observe

- Improvement in speed
- Reduction in critical path delay
- Increase in resource usage

Problem 4: Power Analysis

Task

Perform power analysis of the design

Steps

- Use Vivado power report
- Analyze dynamic and static power

Record

- Total power consumption
- Dynamic power
- Leakage power

Problem 5: Trade-Off Analysis

Task

Compare Area vs Performance vs Power

Table

Mode	LUTs	Registers	Power (mW)	Delay (ns)	Frequency (MHz)
Default					
Area Optimized					
Performance Optimized					

Problem 6: Multiplier Design Trade-Off

Task

Design a **4-bit multiplier** using Verilog.

Steps

- Implement basic combinational multiplier
- Perform synthesis with default settings
- Apply **area optimization** and **performance optimization**

Analyze

- LUT usage (area)
- Critical path delay
- Maximum frequency
- Power consumption

Goal

Compare how multiplier design changes under different optimization strategies.

Problem 7 : Adder Architecture Comparison

Task

Design and compare:

- Ripple Carry Adder (RCA)
- Carry Lookahead Adder (CLA)

Steps

- Implement both in Verilog
- Run synthesis and implementation
- Generate reports

Analyze

- Area (LUTs, registers)
- Delay (critical path)
- Power consumption

Goal

Understand trade-off:

RCA → low area, high delay

CLA → high speed, more area

Analysis Points

- Area optimization reduces hardware but may increase delay
- Performance optimization improves speed but increases area and power
- Power depends on switching activity and resource usage

Evaluation Scheme (10 Marks)

Component	Marks
Verilog Coding	3
Simulation	2
Area/Power Analysis	2
Trade-off Comparison	2
Viva	1